

PRAKRITI DHITAL

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Bachelor of Science in Information Technology student with proven experience building AI-powered systems using Python end-to-end deep learning, custom RAG pipelines, and multi-agent architectures. Strong background in backend development, data workflows, and LLM-powered applications, with an emphasis on reliability, evaluation, and explainability. Seeking entry-level AI or software engineering roles to contribute to responsible, production-oriented AI systems.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, CNNs, RNNs, BiLSTM, Transformers, FFT-MLP, Sensor Fusion, Model evaluation

NLP & AI: Transformer, Positional encoding, Tokenization, RAG pipeline, LangChain, LLM evaluation, Prompt engineering, Multi-Agent systems

Data & ML workflow: Pandas, NumPy, Matplotlib, scikit-learn, EDA, Feature engineering, Train-test evaluation.

Languages & Frameworks: Python, JavaScript, Java, C#, SQL, React, Node.js, ASP.NET MVC

Cloud & Deployment: Google Cloud Run, Docker, Cloud Architecture, RESTful APIs, AWS & Azure familiarity

Tools & Practices: Git, VS Code, Postman, Linux, SQL Server, MySQL, Vector Databases (FAISS, MongoDB)

EDUCATION

Bachelor Of Science in Information Technology

2022-2026

Fairleigh Dickinson University, Vancouver, BC

- Received Academic Honors and Dean List (Spring 2025)
- Relevant Coursework: Neural Networks (AI), Software Engineering, Web Development, Database Systems, OOP.

PROJECTS

Capstone: BFRB Gesture Recognition – Deep learning & NLP *Python, PyTorch, Transformer, BiLSTM, CNN, FFT-MLP*

- Designed and implemented 8 deep learning models — including CNN-BiLSTM, Transformer, BiLSTM, FFT-MLP, and sensor fusion architectures — for 18-class gesture classification on real-world wearable sensor data (IMU, Thermopile, Time-of-Flight).
- Achieved 94.28% Binary F1 (Late Fusion) and 85.98% Macro F1 (Transformer) outperforming all paper baseline models on the 18-class macro metric by 35+ percentage points.
- Developed a novel NLP pipeline converting continuous sensor readings to discrete tokens, enabling Transformer and BiLSTM models to process time-series wearable data as sequential text.
- Conducted full ML lifecycle: EDA, feature engineering, sequence-level train/test split, model training, evaluation (Binary F1, Macro F1, K+1 F1), and reproducibility with fixed random seeds

Local RAG System Document Question Answering (2026) *Python, FAISS, Embeddings, Vector Search, LLMs*

- Built a fully local RAG pipeline for document-based Q&A without external API dependencies, covering ingestion, chunking, embedding, and FAISS vector similarity search.
- Evaluated retrieval quality using similarity search relevance and response grounding techniques to minimize hallucinations.
- Demonstrates applied understanding of LLM architectures, embedding models, and production AI system design.

AI Multi-Agent System – Intelligent Tutor *Python, Flask, Google GenAI(Gemini)SDK, Google Cloud Run, Docker, REST API*

- Designed a 5-agent AI tutoring system (assessment, teaching, practice, curriculum, progress) with intent routing and multi-turn conversational memory using Firestore.
- Deployed to Google Cloud Run with automated build/deploy pipelines, health checks, and containerized Docker architecture.

Network Packet Analyzer (2025) *Python, Scapy, Rich, TCP/IP, JSON, React*

- Built a real-time packet analysis tool parsing Ethernet, IP, TCP/UDP, and ICMP protocols with anomaly detection, statistical tracking, and an interactive web UI.

VOLUNTEERING EXPERIENCE

UN Youth Delegate *United Nations, New York, NY | May 2025*

- Represented youth perspectives in global policy discussions, developing communication and cross-cultural collaboration skills.

Event Coordinator *Fairleigh Dickinson University – Vancouver, BC | Mar 2024 – Present*

- Led technical setups and cross-team coordination for university events, strengthening problem-solving and teamwork skills.